

A decorative background featuring a network diagram. It consists of numerous small circles (nodes) connected by thin lines (edges). Some nodes are solid blue, some are solid grey, and some are outlined in blue. The network is more densely packed on the left and right sides of the slide, with the central area being mostly white space containing the title.

Designing Secure IoT System

Systems Engineering

1. Design standalone device.
2. Understand automation requirements.
3. Identify the desired device software architecture.
4. Identify the desired Edge/Cloud SW architecture.
- 5. Threat modeling -> Mitigation mechanisms.**
6. Design test scenarios.
7. Implement (write code).

Designing Standalone device

Automatic temperature control:

- Identify sensor and actuator.
- Temperature sensitivity.

Insulin Pump:

- Blood sugar sensor and actuator.

Understand automation requirements

- Local controllable:
 - Short range -> 10-100m.
- Remote controllable:
 - Long range -> Around the world.

Identifying the desired device software architecture

- Simple/Fixed functionality:
 - Simple thermostat (display, modify temperature).
 - Simple door lock (lock/unlock/display status)
- Diverse functionality and speed:
 - Robotic Vacuum.
- Fixed but complex functionality:
 - Smart thermostat with HVAC control (display, regulate, schedule, etc)

Identifying the desired device software architecture

- Simple/Fixed functionality (Type 3):
 - Simple thermostat (display, modify temperature).
 - Simple door lock (lock/unlock/display status)
- Diverse functionality and speed (Type 1):
 - Robotic Vacuum.
- Fixed but complex functionality (Type 2):
 - Smart thermostat with HVAC control (display, regulate, schedule, etc)



Identifying the desired Edge/Cloud architecture

- Setup.
- Operational.
- Maintenance.

IoT Device Setup

- Establish ownership.
- Enable Cloud/Hub connection.

IoT Device Setup

- Establish ownership:
 - HSP Property:
 - **H**ave it, **S**ee it and **P**hysically control it.
- Enable Cloud/Hub connection.
 - Configure connection parameters.

IoT Device Setup: Establishing Ownership

- Have it: Can read off a static value from the package/device.
- See it: Can read off a dynamic value from the device.
- Physically control it: Can press a button on the device.



IoT Device Setup: Establishing Ownership

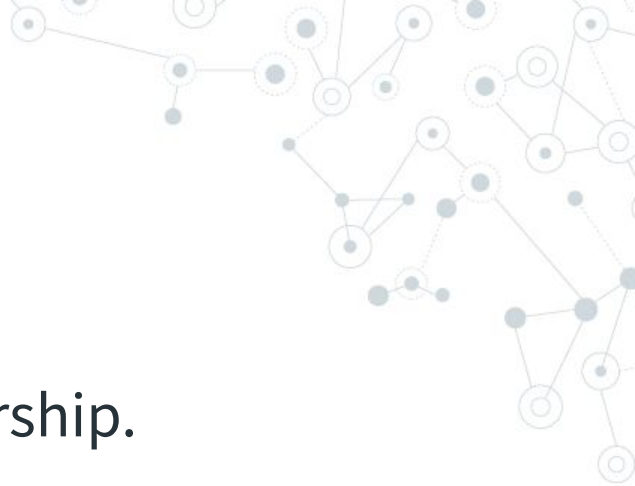
1. “Press a button to enable pairing”. (P)
2. “Smartphone pairs with the device”.
3. “Controller asks the static key”. (H)
4. If successful, “Controller asks a dynamic value displayed on the device”. (S)



IoT Device Setup: Establishing Ownership

1. “Press a button to enable pairing”. (P)
2. “Smartphone pairs with the device”.

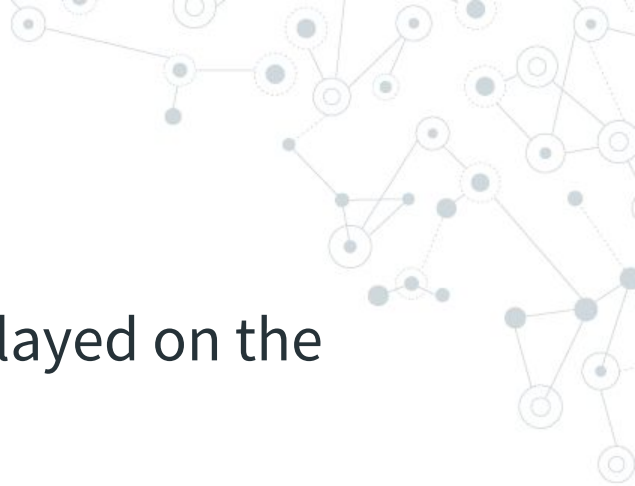
“Attacker could always be in pairing mode”.



IoT Device Setup: Establishing Ownership

1. “Controller asks the static key”. (H)
2. If successful, device establishes ownership.

“Attacker guess/steals the static key”.



IoT Device Setup: Establishing Ownership

1. “Controller asks a dynamic value displayed on the device”. (S)

“Attacker could guess the value” => Our SmartTV work!

IoT Device Maintenance

- Crash recovery:
 - Reboots (robust startup routines).
 - Power failure (proactive warning).
 - Hardware failure (Communicate with user)
- Updates and rollbacks:
 - Ability to update device software and rollback if the update fails.